Euclid's Elements — Book I
Proposition 15

Formal Reconstruction — Technical Documentation

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Vertical angles are equal

1 Introduction

Euclid's Proposition I.15 establishes the equality of vertical angles formed by two intersecting straight lines.

If two straight lines intersect at a point, each pair of opposite angles is equal. In modern terminology, these are the vertical-angle relations.

Euclid proves the proposition by applying Proposition I.13 to two adjacent linear pairs and subtracting a common angle from two sums that are each equal to two right angles.

The Hilbert reconstruction is considerably shorter. The corresponding vertical-angle theorem has already been derived in the Hilbert layer as

`hilbert_vertical_angles`

from Hilbert's theory of adjacent angles. Proposition I.15 therefore becomes an application of an existing synthetic theorem rather than a new geometric derivation.

This proposition provides another example in which a classical Euclidean proof is compressed by the intermediate Hilbert interface: the mathematical content required by Euclid has already been isolated as a reusable theorem.

2 The Classical Configuration

Let the straight lines AB and CD intersect at the point E, with

$$A - E - B$$

and

$$C - E - D.$$

The intersection determines four angles around E.

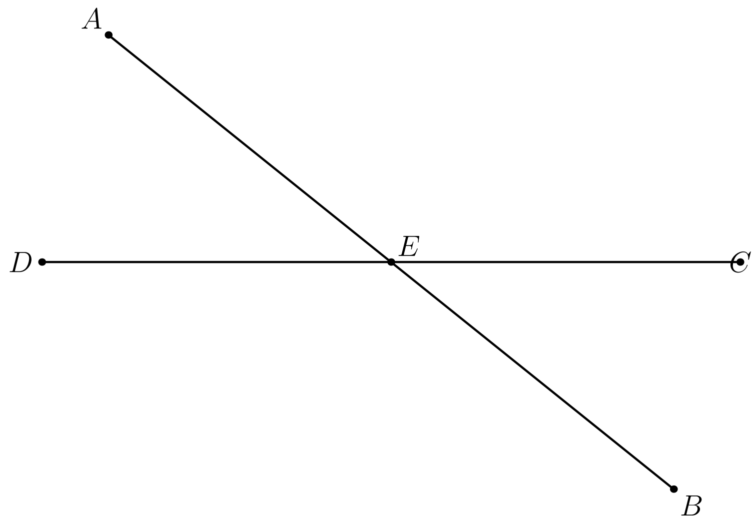

Figure 1: Classical and formal crossing configuration for Proposition I.15. The strict betweenness relations $A - E - B$ and $C - E - D$ encode the two pairs of opposite rays at E.

The opposite pairs are

$$\angle AEC \qquad \text{and} \qquad \angle BED,$$

and

$$\angle CEB \qquad \text{and} \qquad \angle DEA.$$

Proposition I.15 states that each pair is equal.

The betweenness relations

$$A - E - B, \qquad C - E - D$$

express synthetically that EA and EB are opposite rays and that EC and ED are opposite rays. Thus the classical configuration is already naturally adapted to Hilbert's order geometry.

3 Statement

Euclid's Proposition I.15 states:

> If two straight lines cut one another, they make the vertical angles equal to one another.

For the configuration

$$A - E - B, \qquad C - E - D,$$

the proposition gives the two vertical-angle congruences

$$\angle AEC \cong \angle BED$$

and

$$\angle CEB \cong \angle DEA.$$

In the Hilbert reconstruction we assume, in addition, that the two lines are genuinely distinct. This is expressed by

$$\neg\,\mathrm{Collinear}(A, E, C).$$

The formal statement is therefore

$$\begin{array}{l} A - E - B, \\ C - E - D, \\ \neg\,\mathrm{Collinear}(A, E, C) \end{array} \qquad \Longrightarrow \qquad \begin{cases} \angle AEC \cong \angle BED, \\ \angle CEB \cong \angle DEA. \end{cases}$$

The noncollinearity hypothesis is the synthetic expression of a proper intersection of two distinct straight lines. It also guarantees that the angles occurring in the proposition are nondegenerate.

Thus the Lean formulation makes explicit a condition that is implicit in Euclid's phrase "two straight lines cut one another." Strict betweenness specifies the two pairs of opposite rays, while noncollinearity excludes the case in which all five points lie on a single straight line.

4 Classical Proof

Euclid first proves the equality of one pair of vertical angles.

Since the straight line AE stands upon the straight line CD, Proposition I.13 gives

$$\angle CEA + \angle AED$$

equal to two right angles.

Again, since the straight line DE stands upon the straight line AB, Proposition I.13 gives

$$\angle AED + \angle DEB$$

equal to two right angles.

Therefore

$$\angle CEA + \angle AED = \angle AED + \angle DEB.$$

Subtracting the common angle $\angle AED$ from both sides, by Common Notion 3, gives

$$\angle CEA = \angle DEB.$$

Thus one pair of vertical angles is equal.

In the same way,

$$\angle CEB = \angle AED.$$

Therefore, when two straight lines cut one another, they make the vertical angles equal to one another.

5 Proof Architecture of the Reconstruction

The formal reconstruction and Euclid's original proof establish the same geometric fact at different levels of the theory.

Classically, Proposition I.15 is derived from Proposition I.13 twice. Two adjacent linear pairs are each equal to two right angles, and Common Notion 3 removes the common angle. Thus

$$\text{I.13} \longrightarrow \text{two linear-pair equalities} \longrightarrow \text{Common Notion 3} \longrightarrow \text{I.15}.$$

The current Hilbert reconstruction does not reproduce angle addition or subtraction locally. The corresponding result has already been packaged as

`hilbert_vertical_angles`.

The first vertical pair is obtained by a direct call to that theorem. The second pair is obtained from the *same crossing*: Lean reverses the first betweenness relation, transports the required noncollinearity to the reordered triple, and calls `hilbert_vertical_angles` again.

Schematically,

$$\begin{array}{c}
A - E - B, \quad C - E - D, \quad \neg\,\text{Collinear}(A, E, C) \\
\downarrow \\
\texttt{hilbert_vertical_angles} \\
\downarrow \\
\angle AEC \cong \angle BED \\
\text{reorient the same crossing and transport noncollinearity} \\
\downarrow \\
\texttt{hilbert_vertical_angles} \\
\downarrow \\
\angle CEB \cong \angle DEA.
\end{array}$$

The formal proof therefore introduces no new geometric point and no new diagrammatic case. Its extra work is structural: the ordered predicates must be normalized so that the reusable theorem applies in the second orientation.

6 Hilbert Reconstruction

The Hilbert reconstruction starts directly from the synthetic data of the intersection:

$$A - E - B, \qquad C - E - D,$$

together with

$$\neg\,\mathrm{Collinear}(A, E, C).$$

Unlike Euclid's proof, no explicit notion of a sum of angles equal to two right angles is required. The relevant geometric content has already been established in the Hilbert layer as the vertical-angle theorem.

6.1 The First Pair of Vertical Angles

The theorem

```
hilbert_vertical_angles
```

has precisely the form needed for the first pair.

Applied to

$$A - E - B, \qquad C - E - D, \qquad \neg\,\mathrm{Collinear}(A, E, C),$$

it gives directly

$$\angle AEC \cong \angle BED.$$

Thus the first conclusion of Proposition I.15 requires no auxiliary construction.

6.2 The Second Pair of Vertical Angles

For the second pair we view the same intersection with the two straight lines exchanged.

From

$$A - E - B$$

the symmetry of strict betweenness gives

$$B - E - A.$$

We also require the noncollinearity condition

$$\neg\,\mathrm{Collinear}(C, E, B).$$

Suppose, for contradiction, that

$$\mathrm{Collinear}(C, E, B).$$

Since $A - E - B$, the points A, E, B are collinear. Permuting the first collinearity relation and using transitivity of collinearity through the distinct points E and B would then imply

$$\mathrm{Collinear}(A, E, C),$$

contrary to the original hypothesis.

Hence

$$\neg\,\mathrm{Collinear}(C, E, B).$$

We may therefore apply `hilbert_vertical_angles` a second time, now to

$$C - E - D, \qquad B - E - A.$$

This yields

$$\angle CEB \cong \angle DEA.$$

Combining the two applications gives the complete conclusion

$$\angle AEC \cong \angle BED, \qquad \angle CEB \cong \angle DEA.$$

The Hilbert reconstruction is therefore substantially shorter than Euclid's original argument. The classical use of Proposition I.13 and Common Notion 3 has already been absorbed into the previously proved vertical-angle theorem; the only additional formal work is the permutation of the intersection data required for its second application.

7 Lean Formalization

The reconstructed proposition is formalized by the theorem

```
theorem euclid_proposition_15
[HilbertIncidence Geo]
[HilbertCongruence Geo]
(A B C D E : Geo.Point)
(hAEB : Geo.Between A E B)
(hCED : Geo.Between C E D)
(hAEC : not PrimCollinear Geo A E C) :
Geo.AngleCongruent A E C B E D /\
Geo.AngleCongruent C E B D E A := by
```

The conclusion records both pairs of vertical angles:

$$\angle AEC \cong \angle BED$$

and

$$\angle CEB \cong \angle DEA.$$

The first pair is obtained directly:

```
have hFirst :
Geo.AngleCongruent A E C B E D :=
hilbert_vertical_angles
Geo A E C B D
hAEB
hCED
hAEC
```

For the second pair, the first line is read in the opposite direction:

```
have hBEC : Geo.Between B E A :=
(HilbertOrder.between_incidence
A E B hAEB).2.2.2.2
```

The required noncollinearity condition is then derived by contradiction. If C, E, B were collinear, cyclic permutation gives E, B, C collinear. Since A, E, B are already collinear and $E \neq B$, transitivity would imply that A, E, C are collinear, contradicting `hAEC`.

In Lean this is expressed by

```
have hCEB :
not PrimCollinear Geo C E B := by
intro h

have hEBC :
PrimCollinear Geo E B C :=
PrimCollinearCycle Geo C E B h

have hAEBcol :
PrimCollinear Geo A E B :=
(HilbertOrder.between_incidence
A E B hAEB).2.2.2.1

have hEB : E != B :=
(HilbertOrder.between_incidence
A E B hAEB).2.1

have hAECcol :
PrimCollinear Geo A E C :=
```

```
hilbert_primCollinear_trans
Geo A E B C
hEB
hAEBcol
hEBC

exact hAEC hAECcol
```

The second pair of vertical angles is now another direct application:

```
have hSecond :
Geo.AngleCongruent C E B D E A :=
hilbert_vertical_angles
Geo C E B D A
hCED
hBEC
hCEB

exact <hFirst, hSecond>
```

Thus the formal dependency structure is

$$\begin{array}{c}
\texttt{hAEB},\ \texttt{hCED},\ \texttt{hAEC} \\
\downarrow \\
\texttt{hilbert_vertical_angles} \quad \text{(first orientation)} \\
\downarrow \\
\text{betweenness symmetry and collinearity transport} \\
\downarrow \\
\texttt{hilbert_vertical_angles} \quad \text{(second orientation)} \\
\downarrow \\
\angle AEC \cong \angle BED, \qquad \angle CEB \cong \angle DEA.
\end{array}$$

No new geometric axiom or auxiliary geometric construction is introduced in the formalization. Proposition I.15 is obtained entirely from the existing Hilbert incidence, order, collinearity, and vertical-angle theory.

8 Relationship with Earlier Propositions

8.1 Proposition I.13

Proposition I.13 establishes the supplementary-angle configuration created when a straight line stands on another straight line. This is the classical dependency used twice in Euclid's proof of I.15. The current `Proposition15.lean`, however, does not call `euclid_proposition_13`; that earlier content has already been absorbed into the reusable Hilbert vertical-angle theorem.

8.2 Proposition I.14

Proposition I.14 gives the converse straight-line criterion for a supplementary configuration. Together I.13 and I.14 complete the local straight-angle infrastructure around an intersection.

I.15 extracts a different invariant of the same local geometry: opposite, or vertical, angles are congruent. Proposition I.14 is therefore contextual rather than a direct theorem-level dependency of the current I.15 proof.

8.3 Transition to Proposition I.16

Vertical-angle congruence becomes an explicit transport tool in Proposition I.16. The current I.16 production file calls the generic interface theorem `VerticalAngles` directly when it must identify two differently oriented exterior angles. Thus I.15 closes the elementary intersecting-lines block while the same geometric content immediately becomes infrastructure for strict angle comparison.

9 Hilbert and Book Zero Components Used

Proposition I.15 is formally short because its main geometric content has already been isolated in the Hilbert layer.

The proof uses one principal theorem together with elementary order and collinearity machinery required to apply it in the two orientations of the intersection.

9.1 `hilbert_vertical_angles`

The central result is

`hilbert_vertical_angles`

which states that if

$$C - E - B, \qquad D - E - F,$$

and

$$\neg\,\mathrm{Collinear}(C, E, D),$$

then

$$\angle CED \cong \angle BEF.$$

This theorem is the immediate vertical-angle corollary developed in the Hilbert layer after the theorem on adjacent congruent angles.

In Proposition I.15 it is used twice.

The first application gives

$$\angle AEC \cong \angle BED.$$

The second application, after reversing the first straight line and transporting noncollinearity, gives

$$\angle CEB \cong \angle DEA.$$

Thus essentially all of the geometric content of Euclid's Proposition I.15 is already contained in `hilbert_vertical_angles`.

9.2 `HilbertOrder.between_incidence`

The theorem

```
HilbertOrder.between_incidence
```

extracts the elementary order and incidence consequences of strict betweenness.

From

$$A - E - B$$

the proof obtains both

$$B - E - A$$

and the collinearity relation

$$\text{Collinear}(A, E, B).$$

It also supplies the inequality

$$E \neq B,$$

which is needed for the transitivity of collinearity.

Thus a single betweenness hypothesis provides all of the auxiliary order data required for the second application of the vertical-angle theorem.

9.3 `PrimCollinearCycle`

The permutation theorem

```
PrimCollinearCycle
```

cyclically reorders a collinearity relation.

In the contradiction argument,

$$\text{Collinear}(C, E, B)$$

is transformed into

$$\text{Collinear}(E, B, C).$$

This places the points in exactly the orientation required by the transitivity theorem used in the next step.

9.4 `hilbert_primCollinear_trans`

The theorem

```
hilbert_primCollinear_trans
```

expresses the transitivity of collinearity when two distinct common points determine the same line.

In Proposition I.15 we combine

$$\text{Collinear}(A, E, B)$$

with

$$\text{Collinear}(E, B, C)$$

and the inequality

$$E \neq B.$$

The result is

$$\text{Collinear}(A, E, C),$$

contradicting the original noncollinearity hypothesis.

Hence

$$\neg\,\text{Collinear}(C, E, B),$$

which is precisely the hypothesis needed for the second invocation of `hilbert_vertical_angles`.

The complete formal dependency may therefore be summarized as

$$
\begin{array}{c}
A - E - B, \quad C - E - D, \quad \neg\,\mathrm{Collinear}(A, E, C) \\
\downarrow \\
\texttt{hilbert\_vertical\_angles} \\
\downarrow \\
\angle AEC \cong \angle BED
\end{array}
$$

$$
\begin{array}{c}
\text{betweenness symmetry} + \text{collinearity transport} \\
\downarrow \\
\neg\,\mathrm{Collinear}(C, E, B) \\
\downarrow \\
\texttt{hilbert\_vertical\_angles} \\
\downarrow \\
\angle CEB \cong \angle DEA.
\end{array}
$$

No genuinely new Book Zero theorem is required for Proposition I.15. The proposition is therefore primarily an interface theorem over the already established Hilbert vertical-angle theory.

10 Formalization Notes

Proposition I.15 is classically almost immediate from the supplementary-angle theory, but its formal reconstruction highlights a different issue: orientation. At an intersection the geometry is symmetric, whereas the formal predicates are written with ordered point triples. Consequently the proof must recover the appropriate primitive collinearities from betweenness, cycle them into the required orientation, and then invoke the reusable vertical-angle theorem.

The essential pattern is

$$
\begin{array}{c}
\text{betweenness} \\
\downarrow \\
\text{primitive collinearity} \\
\downarrow \\
\text{orientation normalization} \\
\downarrow \\
\texttt{hilbert\_vertical\_angles} \\
\downarrow \\
\text{angle congruence.}
\end{array}
$$

No new geometric point, extension, intersection, or case split is introduced in `Proposition15.lean`. The current source contains one theorem, `euclid_proposition_15`, and its geometric work is delegated to the existing Hilbert interface.

11 Diagrammatic Audit

The preserved Book1R chapter already made the correct diagrammatic choice: a single crossing is sufficient. Figure 1 records the only proposition-specific geometric state,

$$A - E - B, \qquad C - E - D,$$

together with the proper-intersection hypothesis expressed textually by

$$\neg\,\mathrm{Collinear}(A, E, C).$$

The two conclusions

$$\angle AEC \cong \angle BED, \qquad \angle CEB \cong \angle DEA$$

are two readings of the same crossing, not two geometric configurations. Reversing $A - E - B$ to $B - E - A$, cycling collinearity, and deriving $\neg\,\mathrm{Collinear}(C, E, B)$ likewise introduce no new geometry. They are orientation and incidence bookkeeping required by the formal API.

The recovered older TEX contained additional TikZ flowcharts for proof architecture and dependency structure. Those diagrams record logical flow rather than geometry and are not retained as proposition figures. The audited chapter therefore has exactly one Asymptote figure and one panel. A figure records geometry, not tactic state.

12 Dependency Summary

The direct formal dependency structure can be summarized as

$$\begin{gathered} A - E - B, \quad C - E - D, \quad \neg\,\mathrm{Collinear}(A, E, C) \\ \downarrow \\ \texttt{hilbert\_vertical\_angles} \\ \downarrow \\ \angle AEC \cong \angle BED \\ \texttt{HilbertOrder.between\_incidence} \\ + \texttt{PrimCollinearCycle} \\ + \texttt{hilbert\_primCollinear\_trans} \\ \downarrow \\ \neg\,\mathrm{Collinear}(C, E, B) \\ \downarrow \\ \texttt{hilbert\_vertical\_angles} \\ \downarrow \\ \angle CEB \cong \angle DEA \\ \downarrow \\ \texttt{euclid\_proposition\_15}. \end{gathered}$$

The module imports `HilbertBookZero`, but the theorem body does not call Proposition I.13, Proposition I.14, or another proposition-specific Euclidean theorem. The classical dependency on I.13 and Common Notion 3 is represented formally by the already proved generic vertical-angle theorem.

13 Conclusion

Proposition I.15 completes the elementary theory of angles formed by intersecting straight lines. Its classical content is the familiar theorem that vertical angles are congruent. In the current Hilbert reconstruction, essentially all of this geometric content is already contained in `hilbert_vertical_angles`; the local proof mainly manages orientation and collinearity so that this reusable result can be applied twice.

The current `Proposition15.lean` contains no local `sorry` and no local axiom declaration. This documentation audit did not run `lake build` or `#print axioms`, so no fresh compilation or axiomatic-status claim is made here.